



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Fundamentals of ICT Lab 06	Instructor: Ms.Quratulain
Program: AI Fall 2025	Date: 19th November, 2025
Marks: 10	Lab Engineer: Ms. Sibgha Noor
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Instructions: • Submit the file with your names following your registration numbers like AI001_Name. • Copy Paste work will be given no marks.	

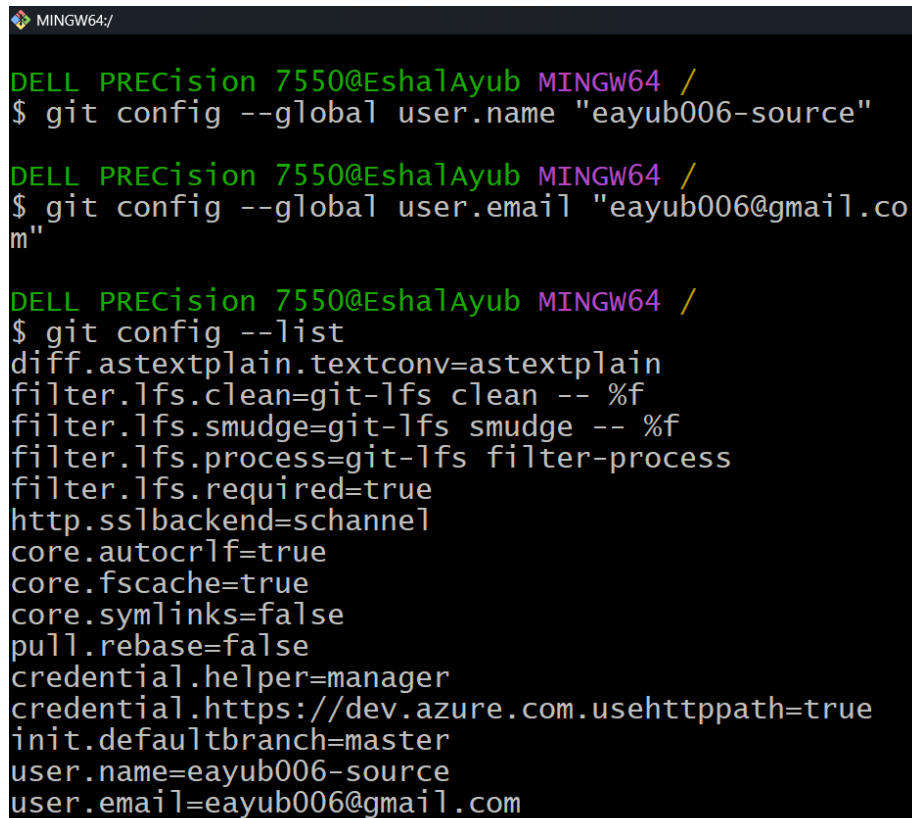
LAB TASK

Lab Learning Objectives

- Learn how to install and configure Git with username and email.
- Understand GitHub account setup and remote repository creation.
- Practice cloning a remote repo to a local system.
- Create files, add, commit, and push using Git commands.
- Modify files and manage version history with Git.

Task 1: Install Git and Configure Username/Email

Screenshot:



```
MINGW64:/  
DELL PRECision 7550@Esha1Ayub MINGW64 /  
$ git config --global user.name "eayub006-source"  
  
DELL PRECision 7550@Esha1Ayub MINGW64 /  
$ git config --global user.email "eayub006@gmail.com"  
  
DELL PRECision 7550@Esha1Ayub MINGW64 /  
$ git config --list  
diff.astextplain.textconv=astextplain  
filter.lfs.clean=git-lfs clean -- %f  
filter.lfs.smudge=git-lfs smudge -- %f  
filter.lfs.process=git-lfs filter-process  
filter.lfs.required=true  
http.sslbackend=schannel  
core.autocrlf=true  
core.fscache=true  
core.symlinks=false  
pull.rebase=false  
credential.helper=manager  
credential.https://dev.azure.com.usehttppath=true  
init.defaultbranch=master  
user.name=eayub006-source  
user.email=eayub006@gmail.com
```

Procedure:

I downloaded Git from the official website and installed it on my system. After installation, I opened Git Bash and set my global username and email using the commands:

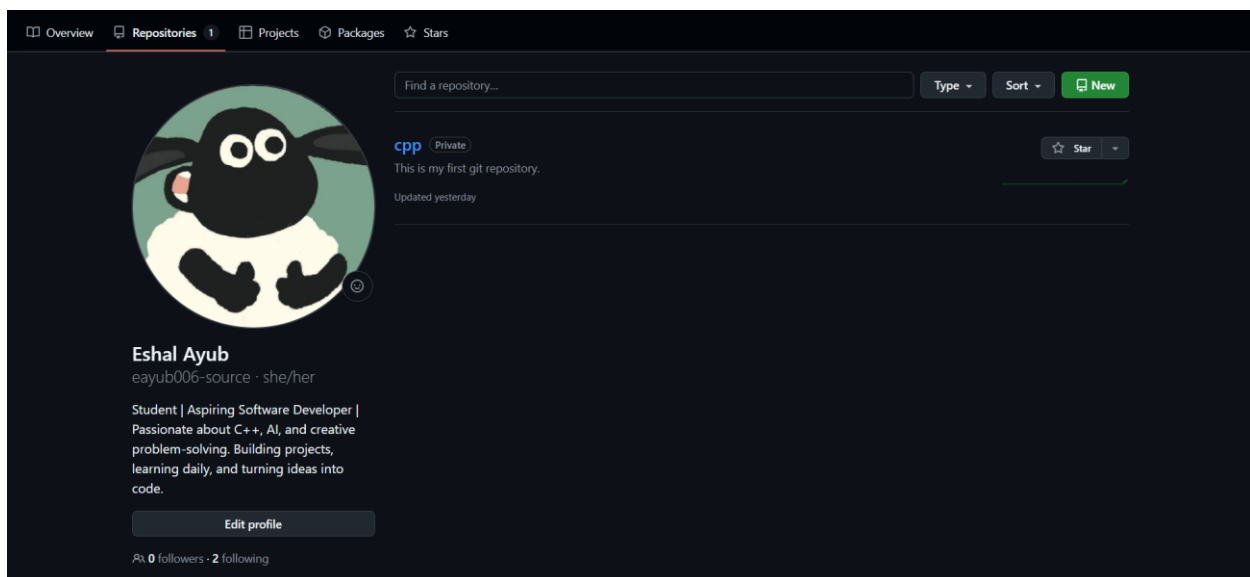
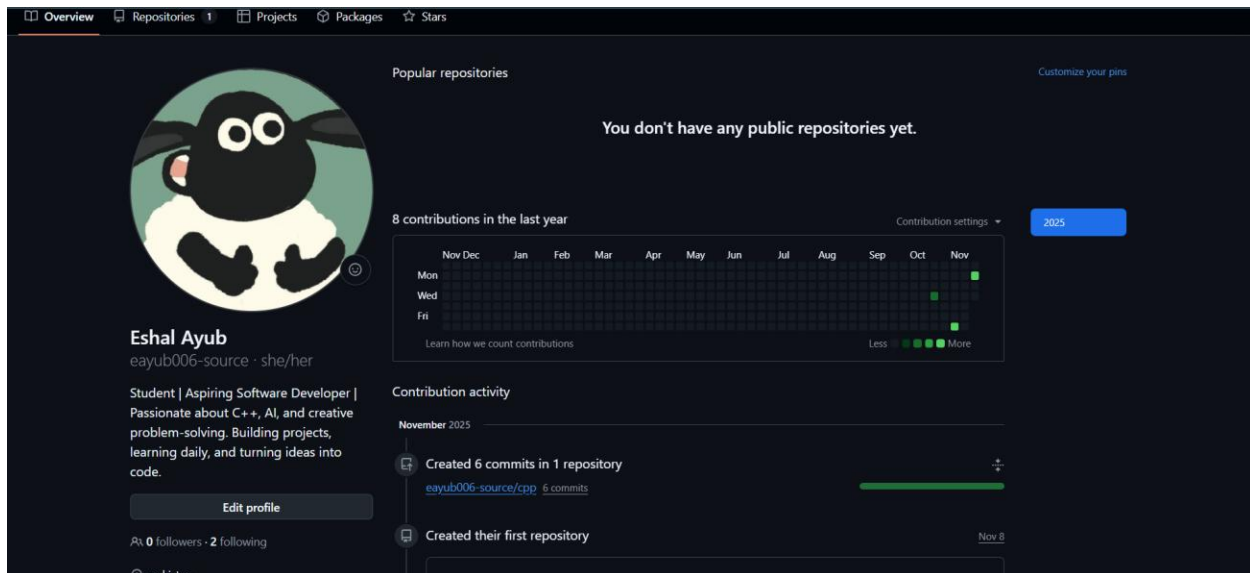
```
git config --global user.name "YourName"
```

```
git config --global user.email "YourEmail".
```

This completed the basic Git setup.

Task 2: Create GitHub Account and New Repository

Screenshot:

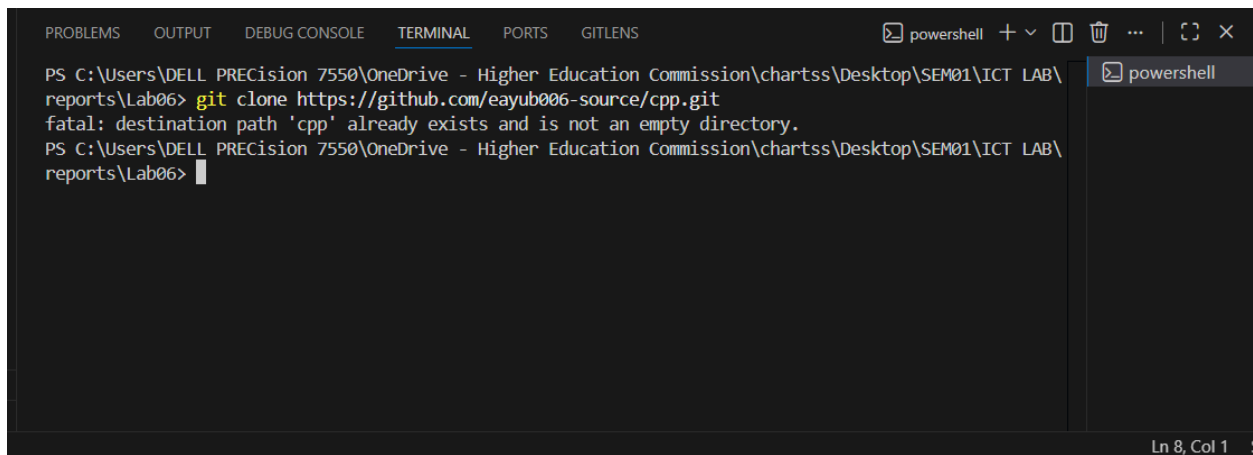


Procedure:

I signed in to GitHub and clicked the option to create a new repository. I named it **cpp**, kept it private, and added no README. After reviewing the settings, I created the repository.

Task 3: Clone Repository to Local System

Screenshot:



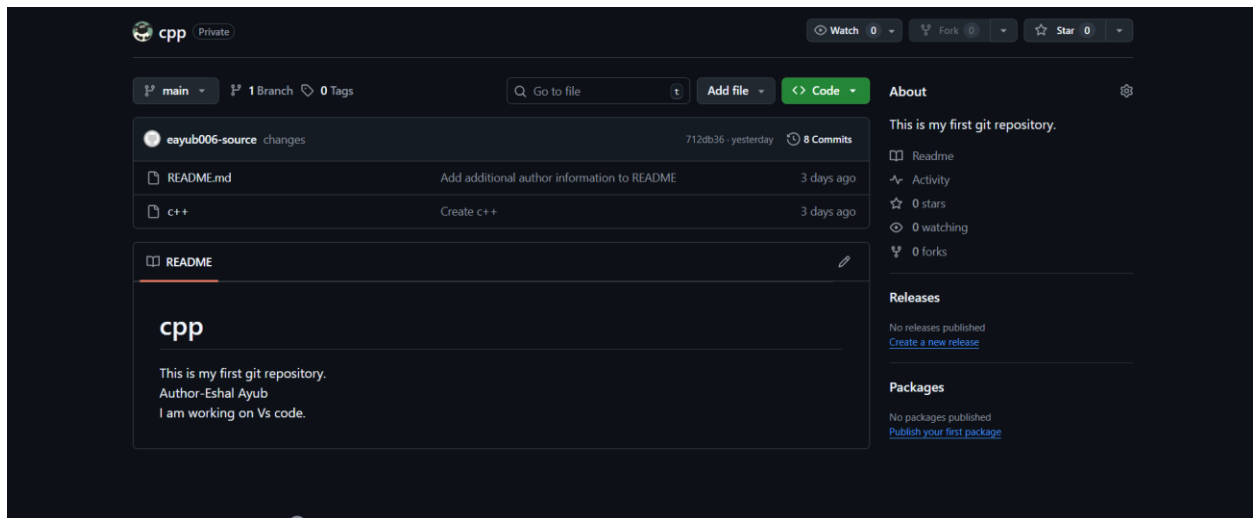
```
PS C:\Users\DELL PRECision 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06> git clone https://github.com/eayub006-source/cpp.git
fatal: destination path 'cpp' already exists and is not an empty directory.
PS C:\Users\DELL PRECision 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06> |
```

Procedure:

I copied the HTTPS link of my GitHub repository. Then I opened Git Bash inside the folder where I wanted to store my project and used the command `git clone <URL>`. This downloaded the repository to my computer.

Task 4: Create a New File and Write a Note

Screenshot:

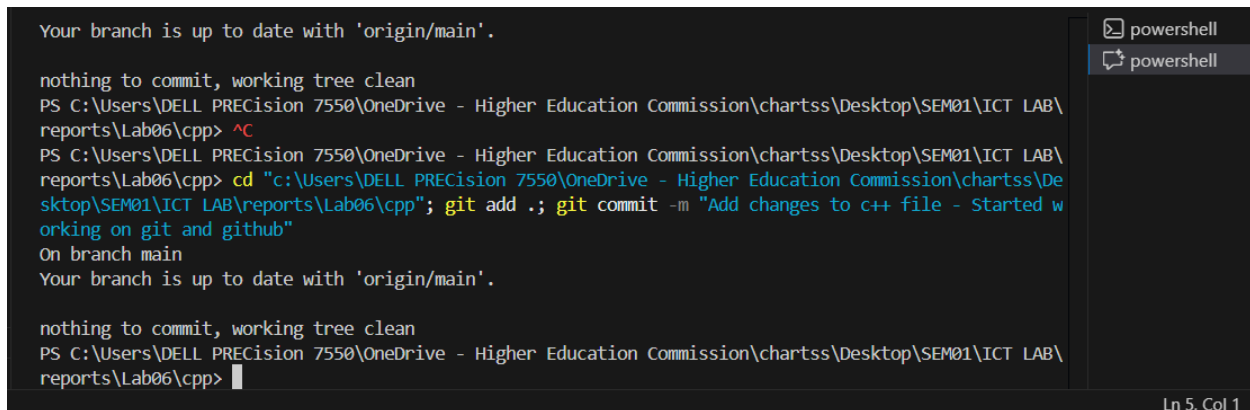


Procedure:

Inside the local repository folder, I created a new file (e.g., note.txt). I wrote a short note explaining what Git and GitHub are. This file was saved directly inside the cloned repository folder.

Task 5: Add and Commit the File

Screenshot:



```
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> ^C
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> cd "c:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp"; git add .; git commit -m "Add changes to c++ file - Started working on git and github"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> 
```

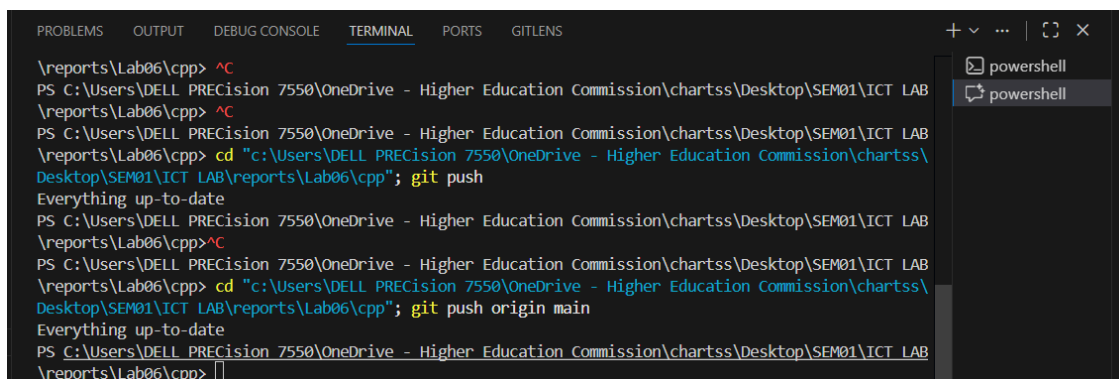
Ln 5, Col 1

Procedure:

I opened Git Bash inside the repository and typed `git add .` to stage the file. Then I committed the changes with the message `git commit -m "Added note about Git and GitHub"`.

Task 6: Push Changes to GitHub

Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

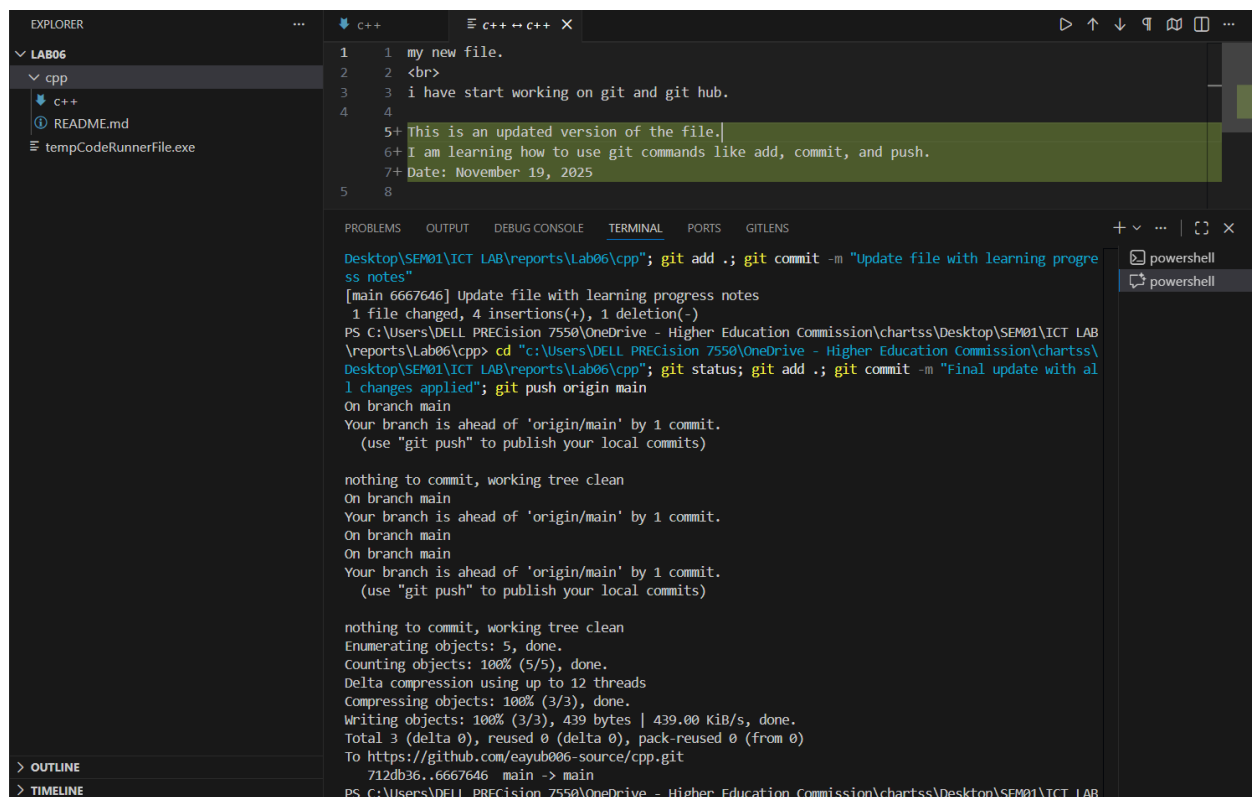
\reports\Lab06\cpp> ^C
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> ^C
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> cd "c:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp"; git push
Everything up-to-date
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> ^C
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> cd "c:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp"; git push origin main
Everything up-to-date
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> 
```

Procedure:

To upload my commit to GitHub, I used the command `git push origin main`. After the push finished, I checked GitHub and confirmed the new file appeared in the repository.

Task 7: Modify the File and Commit Again

Screenshot:



The screenshot shows the Visual Studio Code interface. On the left, the Explorer pane shows a project named 'LAB06' with a 'cpp' folder containing 'README.md' and 'tempCodeRunnerFile.exe'. The main editor shows a C++ file with the following content:

```
1 my new file.
2 <br>
3 i have start working on git and git hub.
4
5+ This is an updated version of the file.
6+ I am learning how to use git commands like add, commit, and push.
7+ Date: November 19, 2025
8
```

The bottom panel shows the TERMINAL window with the following output:

```
Desktop\SEM01\ICT LAB\reports\Lab06\cpp"; git add .; git commit -m "Update file with learning progress notes"
[main 6667646] Update file with learning progress notes
1 file changed, 4 insertions(+), 1 deletion(-)
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp> cd "c:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB\reports\Lab06\cpp"; git status; git add .; git commit -m "Final update with all changes applied"; git push origin main
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
On branch main
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 439 bytes | 439.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eayub806-source/cpp.git
712db36..6667646 main -> main
PS C:\Users\DELL PRECISION 7550\OneDrive - Higher Education Commission\chartss\Desktop\SEM01\ICT LAB
```

Procedure:

I opened the same file again and added more lines to the note. After saving, I checked the status using `git status`. I added the updated file using `git add .`, committed with a new message, and pushed the changes again to GitHub.

Summary / Conclusion

This lab helped me understand how version control works step-by-step. I learned how to install Git, connect it with GitHub, and manage files using basic commands like clone, add, commit, and push. The easiest part was creating the repository, while the most challenging part was remembering the correct Git commands. Overall, the lab improved my confidence in using version control for future projects.